

Tuning Oscillator Frequency and Learning About Poles

Anderson Hsieh

July 19, 2026

1 Questions Answered

1. How do I sweep a variable in xschem & ngspice and run transient analysis for each value of such variable?

Do this :

```
.param temp=127
.param Vtune = 0
.control

let sweep_start = 0.1
let sweep_stop = 1.2
let sweep_step = 0.1
let sweep_value = sweep_start

while sweep_value <= sweep_stop
  alterparam Vtune = ${sweep_value}
  reset

  save v(out)
  tran 5p 20n

  linearize v(out)
  set specwindow=blackman
  fft v(out)
  write ro3_tran_fft_${sweep_value}.raw v(out)

  let sweep_value = sweep_value + sweep_step
end
.endc
```

2 Questions To Be Answered

1. how do they tune VDD for ring oscillator in a real application?
2. poles and zeros in oscillators
3. current mirrors in differential ring oscillator

3 Some Picture

